

L35: Validation Evidence

Simulation Without a Black Box

Outline

Types of Validation

Historical Validation

Quantitative Validation Metrics

Structured Walkthrough

Predictive and Sensitivity Validation

Partial Validation

Good-Enough Criteria

Key Takeaways

Three Types of Validation

Face Validity

Does the model *look* reasonable to domain experts?

Method: structured walkthrough with practitioners.

Internal Validity

Are results consistent across replications? Is variance explained by randomness, not logic errors?

Method: replicate widely; check CIs.

External / Historical

Do simulated outputs match observed historical data from the real system?

Method: compare KPIs to records.

All three are necessary; none is sufficient alone.

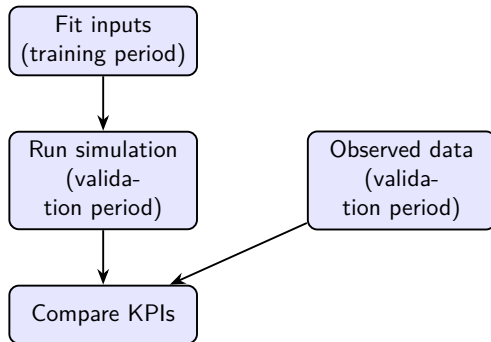
Historical Validation: Comparing to Observed Data

Procedure:

1. Collect historical data: arrival logs, service durations, timestamps.
2. Fit input distributions to the historical record.
3. Run the simulation over the *same* operating period.
4. Compare simulated KPIs to observed KPIs using a two-sample *t*-test or confidence-interval overlap.

Warning

Do not use the same data set to both *fit inputs* and *validate outputs* — this is circular validation. Use a held-out validation period.



Quantitative Metrics: Theil's U and RMSE

Root Mean Square Error over n matched observations:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i^{\text{sim}} - y_i^{\text{obs}})^2}$$

Theil's Inequality Coefficient:

$$U = \frac{\sqrt{\frac{1}{n} \sum (y_i^{\text{sim}} - y_i^{\text{obs}})^2}}{\sqrt{\frac{1}{n} \sum (y_i^{\text{sim}})^2} + \sqrt{\frac{1}{n} \sum (y_i^{\text{obs}})^2}}$$

$U = 0$: perfect fit. $U = 1$: naive forecast is just as good.

When to Use Each

- **RMSE**: same units as the KPI; interpretable as average error.
- **Theil's U**: dimensionless; useful for comparing models with different KPI scales.
- **CI overlap**: preferred when replications are available — respects simulation variance.

Structured Walkthrough: Face Validity

Process:

1. Assemble domain experts (clinicians, operations staff, managers).
2. Walk through the conceptual model diagram *event by event*.
3. Ask: “Does this match how the real system works?”
4. Record every objection — each is either a model fix or a documented limitation.
5. Repeat after significant model changes.

What Experts Check

- Are all patient types / entity classes represented?
- Are routing probabilities plausible?
- Are service-time distributions in the right range?
- Is the priority logic correct?

Warning

Expert approval is not validation — it is face validity only. Always follow up with data.

Predictive and Sensitivity Validation

Predictive Validation

Train on historical period $[0, T_1]$. Predict $[T_1, T_2]$. Compare predictions to observations in $[T_1, T_2]$ before revealing them.

Strongest form of validation — model made a blind prediction.

Sensitivity Validation

Do outputs respond to inputs in *plausible directions*?

- Increasing λ should increase W .
- Adding a server should decrease W_q .
- Reducing lead time should cut average inventory.

If a sensitivity direction is wrong, there is a structural model error.

Partial Validation: When Perfect Data Is Unavailable

Reality: complete historical records rarely exist.

Partial validation strategy:

- Validate the sub-models where data is available (e.g., arrival process).
- Use face validity for components without data (e.g., rare events).
- Document explicitly what was validated, what was assumed.
- Quantify the uncertainty added by unvalidated components via sensitivity analysis.

Documentation Template

Component	Evidence
Arrivals	Historical log
Service times	Time study
Routing	Expert estimate
Priority logic	Face valid

Every row must appear in the report.

When Is a Model “Good Enough”?

Credibility is decision-relative, not absolute.

- **Low-stakes decision** (e.g., scheduling shift): face validity + one historical comparison may suffice.
- **High-stakes decision** (e.g., \$10M capital investment): historical + predictive validation + sensitivity analysis required.
- **Policy recommendation**: document scope of validity — the model is only valid for the system and conditions tested.

Common Mistake

Reporting a validated model without stating the *scope* of validation. A model validated for $\rho \in [0.5, 0.8]$ should not be used to predict $\rho = 0.99$ behavior without additional validation.

Rule of Thumb

The required validation effort scales with the cost of a wrong decision.

Key Takeaways

L35 Summary

1. Validation has three types: face, internal, and historical/external — all are needed.
2. Historical validation requires a *held-out* period; circular validation (fit and validate on same data) proves nothing.
3. RMSE and Theil's U quantify fit; CI overlap is preferred when replications exist.
4. Sensitivity validation checks plausibility of input–output direction.
5. Good-enough is decision-relative — state the scope of validity explicitly in every report.

Next: L36 — Model credibility, documentation, and reproducibility.